Desk Buddy

Project Progress: Problem and Design

IoT and Applications

Hanoi University of Science and Technology

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The problem

A long session at a closed desk

- Heat, humidity, dim light, heavy air
- Easy to miss until focus drops
- Breaks are forgotten

Desk Buddy

One node for one study desk

- Senses the desk environment
- Sends one reading to the laptop
- Shows live numbers and history
- Beeps when a break is due

Scope of this talk

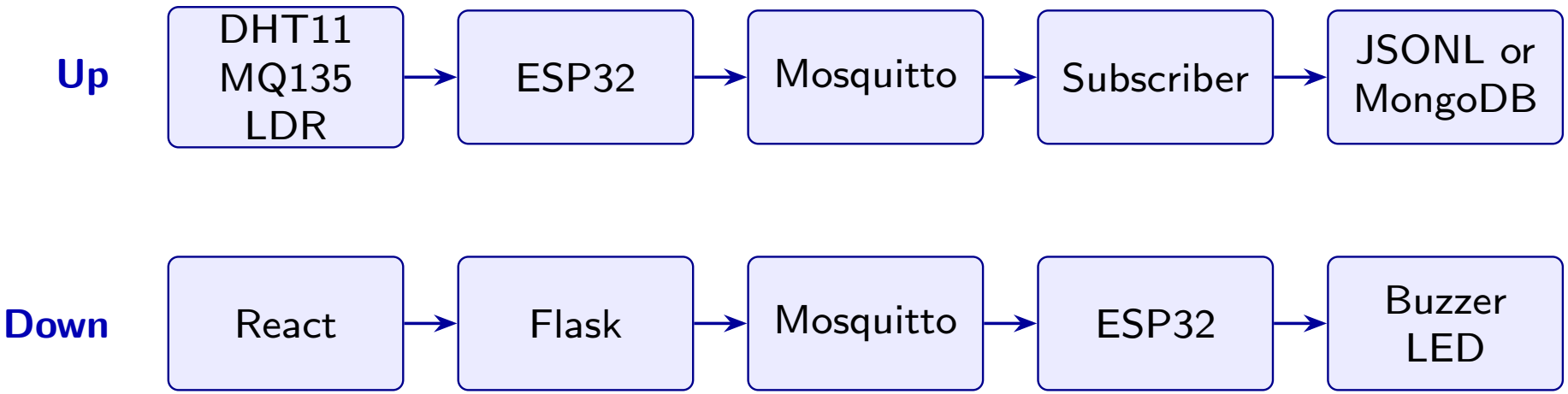
In scope

- One node: `node1`
- One JSON every 5 seconds
- Sensors, MQTT, and a web page

This report

- Problem, architecture, and choices
- Build and measurements come next

Overall architecture



The browser talks to Flask only.

Message and thresholds

Every 5 seconds

Field	Meaning
`tempC`	DHT11, °C
`humidity`	DHT11, %
`airRaw`	MQ135, ADC
`lightRaw`	LDR, ADC

Alert when

Check	Limit
Hot	$\geq 32°C$
Humid	$\geq 80\%$
Air	≥ 1800
Dim	≤ 400

ADC stays 0–4095.

No ppm, no lux.

Choosing the board

Choice	Why	Alternative
ESP32	Wi-Fi on the chip. ADC1 still reads while Wi-Fi is on.	Arduino Uno: no Wi-Fi. ESP8266: one ADC. Raspberry Pi: too heavy for one node.

Air and light use GPIO 34 and 35.

Choosing the sensors

Measure	**Choice**	**Why**	**Alternative**
Heat, humidity	**DHT11**	Enough for 32°C and 80%	DHT22, BME280
Air	**MQ135, raw ADC**	Higher means worse air	Digital ppm sensor
Light	**LDR, raw ADC**	Brighter means a higher value	BH1750 in lux
Desk alert	**Active buzzer and LED**	A high pin turns it on	Passive buzzer

Choosing the software

Layer	Choice	Why	Alternative
Link	**MQTT, two topics**	Readings up, commands down	HTTP, CoAP
Broker	**Mosquitto on the laptop**	Runs in the lab	Cloud broker
Services	**Subscriber and Flask**	The web uses HTTP only	Board writes the database
Store	**JSONL now, MongoDB later**	Demo still runs offline	MongoDB only
UI	**React dashboard**	Live view in the browser	Serial monitor

Where we are

Started

- Contract is fixed: topics, fields, limits
- Laptop path runs with the simulator

Next

- Flash the firmware
- Read the real sensors
- Fill the measurement table

Summary

- One desk node for comfort and breaks
- ESP32, DHT11, MQ135, LDR
- MQTT on the laptop, then Flask and React

Thank you for listening!!